DIF2708

$x^2 = 2 \cdot x^{(x^3)} = 3x^2 \cdot$

1.

$\operatorname{sen}(A)\cos(A) = \frac{1}{2}\operatorname{sen}(2A)$

$\cos^2(\alpha) = \frac{1}{2} + \frac{\cos(2x)}{2}$

$\operatorname{sen}^2(\alpha) = \frac{1}{2} - \frac{\cos(2x)}{2}$

$\operatorname{sen}^2(\alpha) + \cos^2(\alpha) = 1$

$\tan^2(\alpha) + 1 = \sec^2(\alpha)$

a) $\int \operatorname{sen}^2(7x)\,dx =$ [illegible]

$= \int \left(\frac{1}{2} + -\frac{\cos(14x)}{2}\right) dx$

$= \int \frac{1}{2}dx - \frac{1}{2}\int \cos(14x)\,dx = \frac{x}{2} - \frac{1}{2}\left(\frac{\sin(14x)}{14}\right) + C$

b) $\int \cos^2(9x)\,dx = \int \left(\frac{1}{2} + \frac{\cos(18x)}{2}\right) = \int \frac{1}{2}dx + \frac{1}{2}\int \cos(18x)\,dx$

$\left(\frac{x}{2}\right)' = \frac{1}{2}dx$

$= \frac{x}{2} + \frac{1}{2}\left(\frac{\operatorname{sen}(14x)}{14}\right) + C \quad \left(\frac{\sin(14x)}{14}\right)' = \frac{1}{14}\left(\sin(14x)\right)' = \frac{1}{14}\cdot 14\cos(14x) = \cos(14x)$

c) $\int \operatorname{sen}(x)\cos(x)\,dx = \int u\,du = \frac{u^2}{2}$

$u = \operatorname{sen}(x)$

$du = \cos(x)\,dx$

$= \frac{\operatorname{sen}^2(x)}{2} + C$

d) $\int \operatorname{sen}(5x)\cos(5x)\,dx = \frac{1}{5}\int u\,du = \frac{1}{5}\left(\frac{u^2}{2}\right)$

$u = \operatorname{sen}(5x)$

$du = 5\cos(5x)$

$\frac{du}{5} = \cos(5x)$

$= \frac{\operatorname{sen}^2(5x)}{10} + C$

e) $\int \cos(7x)\cos(7x)\,dx = \int \cos^2(7x)\,dx = \int \left(\frac{1}{2} + \frac{\cos(14x)}{2}\right) dx = \frac{1}{2}\int \cos(14x) + \frac{x}{2}$

$= \frac{\sin(14x)}{28} + \frac{x}{2} + C$

f) $\int \cos^3(x)\,dx = \int \cos^2(x)\cos(x)\,dx$

$= \int (1 - \operatorname{sen}^2(x))\cos(x)\,dx = -\int (\operatorname{sen}^2(x) - 1)\cos(x)\,dx = -\int (u^2 - 1)\,du = -\left(\frac{u^3}{3}\right) + C$

$u = \operatorname{sen}(x)$

$du = \cos(x)\,dx$

$= \frac{-\operatorname{sen}^3(x)}{3} + C$

g) $\int \tan^2(x)\,dx = \int (\sec^2(x) - 1)\,dx = \int \sec^2(x)\,dx - \int 1\,dx = \tan(x) - x + C$

h) $\int (1 + \tan^2(x))\sec(x)\,dx = \int \sec^2(x)\sec(x)\,dx$

$= \int \sec(x)\,dx + \int \sec(x)\tan^2(x)\,dx = \int \sec(x)(\sec^2(x) - 1)\,dx$

$= \ln|\sec(x) + \tan(x)| + \int \sec(x)(\sec^2 x - 1)\,dx = \int \sec^3(x) - \int \sec(x)$

$= \int \sec^3(x) + \ln|\sec(x) + \tan(x)| - \int \sec(x) = \ln|\sec(x) + \tan(x)|$

h) $\int (1 + \tan^2(x))\sec^2(x)\,dx = \int \sec^2(x)\,dx + \int \sec^2(x)\tan^2(x)\,dx$

$= \tan(x) + \int u^2\,du$ $\quad u = \tan$, $du = \sec^2(x)\,dx$

$= \tan(x) + \frac{u^3}{3} = \tan(x) + \frac{\tan^3(x)}{3} + C$

2. $\int \sin^5(x)\cos^4(x)\,dx =$

$\int \sin^4(x)\sin(x)\cos^4(x)\,dx = \int (1 - \cos^2(x))^2 \sin(x)\cos^4(x)\,dx$

$u = \cos(x)$

$du = \sin(x)\,dx$

$= \int (1 - u^2)^2 u^4\,du$

$= \int (u^2 - 1)^2 u^4\,du = -\int (u^4 - 2u^2 + 1)u^4\,du = \int u^8\,dx - 2\int u^6\,dx + \int u^4\,dx$

$= \frac{u^9}{9} - 2\left(\frac{u^7}{7}\right) + \frac{u^5}{5} = \frac{\cos^9(x)}{9} - \frac{2}{7}\cos^7(x) + \frac{\cos^5(x)}{5}$

2* a)* $\int \operatorname{sen}^5(x)\cos^4(x)\,dx = \int \operatorname{sen}(x)\operatorname{sen}^4(x)\cos^4(x)\,dx$

$= \int \sin(x)(\cos^2(x)-1)^2\cos^4(x)\,dx =$

$u = \cos(x)$

$du = -\sin(x)$

$-du = \sin(x)\,dx$

$= \int -(u^2-1)^2 u^4\,du$

$= -\int (u^4-2u^2-1)u^4\,du$

$= -\int u^8\,du + 2\int u^6\,du + \int u^4\,du =$

$\frac{-u^9}{9} + \frac{2u^7}{7} + \frac{u^5}{5} = \frac{-\cos(x)}{9} + \frac{2\cos^7(x)}{7} + \frac{\cos^5(x)}{5} + C$

e) $\int \operatorname{sen}^3(x)\,dx = \int \sin(x)\operatorname{sen}^2(x)\,dx$

$= \int \sin(x)(\cos^2-1)\,dx = \int (u^2-1)\,du$

$u = \cos(x)$

$du = -\sin(x)\,dx$

$= \int u^2\,du - \int 1\,du = \frac{u^3}{3} - x = \frac{\cos^3(x)}{3} - x$

f) $\int \cos^4(x)\,dx = \int \cos^2(x)\cos^2(x)\,dx = \int \left(\frac{1}{2} + \frac{\cos(2x)}{2}\right)^2 dx$

$= \int \frac{1}{4} + \frac{\cos(2x)}{2} + \frac{\cos^2(2x)}{2}\,dx = \frac{1}{4}\int dx + \frac{1}{2}\int \cos(2x)\,dx + \int \frac{\frac{1}{2} + \frac{\cos(4x)}{2}}{2}\,dx$

$= \frac{x}{4} + \frac{\sin(2x)}{4} + \frac{x}{4} + \frac{1}{4}\int \cos(4x)\,dx = \frac{x}{2} + \frac{\operatorname{sen}(2x)}{4} + \frac{\cos(4x)}{16} + C$

b) $\int sen^4(3x)\cos^5(3x)\,dx = \int sen^4(3x)\cos(3x)\cos^4(3x)\,dx$

$= \int sen^4(3x)\cos(3x)(\sin^2(3x)-1)^2\,dx = \int u^4(u^2-1)^2\,du$

$u = \sin(3x)$
$du = 3\cos(3x)$
$\frac{du}{3} = \cos(3x)$

$= \int u^4(u^4-2u^2+1)\,du = \int u^8\,du - 2\int u^6\,du + \int u^4\,du$

$= \frac{u^9}{9} - 2\left(\frac{u^7}{7}\right) + \frac{u^5}{5}$

$= \frac{\sin^9(3x)}{9} - \frac{2}{7}\sin^7(3x) + \frac{sen^5(3x)}{5}$

c) $\int sen^8(x)\cos^3(x)\,dx = \int sen^8(x)\cos^2(x)\cos(x)\,dx = \int \sin^8(x)(\sin^2(x)-1)\cos(x)\,dx$

$u = \sin(x)$
$du = \cos(x)\,dx$

$= \int u^8(u^2-1)\,du = \int u^{10}\,du - \int u^8\,du = \sin^{10}(x) - \sin^8(x)$

d) $\int sen^2(6x)\cos^9(6x)\,dx = \int sen^2(6x)\cos^8(6x)\cos(6x)\,dx = \int sen^2(6x)(\sin^2(6x)-1)^4\cos(6x)\,dx$

$u = \sin(6x)$
$du = 6\cos(6x)\,dx$
$\frac{du}{6} = \cos(6x)\,dx$

$= \frac{1}{6}\int u^2(u^2-1)^4\,du = \frac{1}{6}\int u^2(u^2-1)^2(u^2-2u^2+1)\,du$

$= \frac{1}{6}\int u^2(u^2-2u^2+1)^2\,du = \frac{1}{6}\int u^2(u^2-2u^4-2u^2-2u^4\ldots)\,du$

$u^2(u^2-2u^2+1)(u^2-2u^2+1)$
$u^4 - 2u^4 + u^3 - 2u^4 + 4u^4 - 2u^3$
$+u^2 - 2u^3 + 1$

$= \frac{1}{6}\int u^2(u^4-2u^2+1)\,du = \frac{1}{6}\int u^6 - 2u^3 + u^2$

$= \frac{1}{6}\int u^6\,du - \frac{1}{3}\int u^3\,du + \frac{1}{6}\int u^2 = \frac{u^7}{42} - \frac{u^4}{12} + \frac{u^3}{18} = \frac{\sin^7(6x)}{42} - \frac{\sin^4(6x)}{12} + \frac{\sin^3(6x)}{18} + C$